

Investigations in Number, Data, and Space® © 2012 for the Common Core: Product Implementation Essentials

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PARTICIPANT WORKBOOK

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Product Implementation Essentials Fused Online Course
Participant Workbook

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Product Implementation Essentials, Participant Workbook**

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Agenda

Session	Agenda Items
1: Investigations and the Common Core State Standards for Mathematics	Introduction • Course Overview • Course Organization • Icebreaker <i>Activity: Remembering Elementary School Mathematics Poll</i> Content Focus • Session Objectives • Overview of the CCSSM <i>Whiteboard Activity: What the CCSSM Are and Are Not</i> • Standards for Mathematical Content <i>Chat Pod Activity: Standards for Mathematical Content</i> <i>Whiteboard Activity: Standards, Clusters, and Domains</i> • Investigations and the Common Core State Standards Resource Book • The Standards for Mathematical Practice • Analyzing the Mathematical Practices <i>Activity: Analyzing the Mathematical Practices</i> <i>True-False Poll: Mathematical Practices</i> • Student Centered Learning • Teacher-Student-Curriculum Partnership <i>Activity: Polling Problem</i> <i>Activity: Solve a Higher-Level Thinking Task</i> • Thoughts on the Standards for Mathematical Practice – Video: Phil Daro: Introduction to the Standards for Mathematical Practices Session 1 Review Prepare for Next Session • Review list of assignments to complete prior to Session 2 • Discussion Board • Course Navigation Mini-Tutorial • Closing

Agenda

Session	Agenda Items
2: A Closer Look at Investigations Curriculum Units	Introduction • Welcome • Session 1 Review • Session 2 Objectives • Background Learn • Structure of a Curriculum Unit • Your Math Day • Explore a Session? <i>Activity: Explore a Session?</i> • Professional Development • Teacher Notes • Dialogue Boxes • Pause and Reflect • Differentiated Instruction • Supporting the Range of Learners – Video: Supporting the Range of Learners • Differentiation and Intervention Guide – Video: Differentiation in Investigations • Quick Check In Practice • Teaching an Investigations Session – Video: Teaching an Investigations Session <i>Activity: Video Reflection</i> Apply • In the Classroom • Assignment • Discussion Board Review • Review Topics Covered in Session 2 Prepare for Next Session • Review Assignments and Discussion Board Questions • Assessment for Learning

Agenda

Session	Agenda Items
3: Building a Mathematical Community of Learners	Introduction • Welcome • Session 2 Review • Session 3 Objectives • Background Learn • Building a Mathematical Community • Mathematical Discussions – Video: Mathematical Discussions • Rationale for Games <i>Activity: Play a Game</i> • Pause and Reflect • The Role of Games • Computational Fluency <i>Activity: Computational Fluency</i> • Strategic Thinking • Supporting the Range of Learners • Quick Check In Practice • Games in the Classroom • Video: Games in the Classroom Apply • In the Classroom • Assignment • Discussion Board Review • Review Topics Covered in Session 3 Prepare for Next Session • Review Assignments and Discussion Board Questions

Agenda

Session	Agenda Items
4: Assessment in Investigations	Introduction • Welcome • Session 3 Review • Session 4 Objectives • Background Learn • How do you assess? • Where is the assessment? • Opportunities for Assessment <i>Activity: Opportunities for Assessment</i> • Ongoing Assessments • Writing Opportunities • Portfolio Opportunities • Assessing the Benchmarks – Video: Assessing the Benchmarks • Pause and Reflect • End-of-Unit Assessment • Classroom Example • Quick Check In Practice • Ongoing Assessment – Video: Ongoing Assessment <i>Activity: Video Reflection</i> Apply • In the Classroom • Assignment • Discussion Board <i>Review</i> • Review Topics Covered in Session 4 Prepare for Next Session • Review Assignments and Discussion Board Questions

Agenda

Session	Agenda Items
5: Planning an Investigations Unit	Introduction - Welcome - Introductions Course Review - Review Course Outcomes - Review Sessions 1–4 <i>Activity: Poll Course Reviews: Sessions 1–4</i> - Open Q&A - Preview Session 5 - Session 5 Objectives Content Focus - Unit Planning <i>Activity: Investigations Program Components Poll</i> - Unit Planning with the CCSSM - Investigations for the Interactive Whiteboard <i>Video: Investigations for the Interactive Whiteboard</i> - Investigations Lesson Debrief <i>Activity: Investigations Lesson Debrief in Breakout Rooms</i> - Peer Feedback Review - Session 5 Review Closing - Review Course Outcomes <i>Activity: 1. . . 2. . . 3</i> - Final Discussion Board - Review Facilitator Contact Information

Outcomes

At the conclusion of this course, you will be able to

- demonstrate an understanding of the instructional philosophy, mathematical content, lesson structure, differentiation, and assessment features of the Investigations curriculum;
- identify the organization and content of the curriculum and supporting resources; and
- describe effective strategies for successfully implementing Investigations into the classroom.

Course Assignment and Preparation Checklist

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	Session Preparation <i>Complete before attending the session.</i>		Session Assignments <i>Complete after attending the session.</i>	
1: Investigations and the Common Core State Standards for Mathematics	Bring your Core Curriculum Package to the first session.		1. Post your response to the Discussion Board. 2. Respond to two Discussion Board posts from other participants. 3. Choose an Investigations session for your final assignment—the Lesson Plan Implementation.	
2: A Closer Look at Investigations Curriculum Units	1. Read pages 8–9 in your <i>Investigations and the Common Core State Standards</i> resource book. 2. Review the curriculum resources in your Core Curriculum Package		1. Complete the Session 2 assignment: Differentiation: Supporting the Range of Learners. 2. Continue filling out the Lesson Plan Implementation organizer as you develop new understandings about implementing Investigations. 3. Post your response to the Discussion Board. 4. Respond to two Discussion Board posts from other participants.	
3: Building a Mathematical Community of Learners	Read the article, “Classroom Influences on Learning.”		1. Complete the Session 3 assignment: Making Math Discussions Explicit. 2. Continue filling out the Lesson Plan Implementation organizer as you develop new understandings about implementing Investigations. 3. Post your response to the Discussion Board. 4. Respond to two Discussion Board posts from other participants.	

Course Assignment and Preparation Checklist

	Session Preparation <i>Complete before attending the session.</i>		Session Assignments <i>Complete after attending the session.</i>	
4: Assessment in Investigations	Read the article, "Assessing to Learn and Learning to Assess."		1. Complete the Session 4 assignment: Assessing Students' Understanding. 2. Post your response to the Discussion Board. 3. Respond to two Discussion Board posts from other participants. 4. Teach the Investigations lesson that you have been focusing on in your final assignment. 5. Complete the final assignment by filling in all parts of the Lesson Plan Implementation graphic organizer and uploading it to the dropbox for this course on Pearson LearningStudio.	
5: Planning an Investigations Unit	Consider possible topics of discussion that relate to your Investigations Lesson Plan Implementation to share during a Lesson Study group discussion.		1. Complete the 3-2-1 assignment. 2. Respond to the final Discussion Board questions. 3. Respond to the responses of two other participants.	

Session 1: Investigations and the Common Core State Standards for Mathematics

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Session 1 Objectives

By the end of Session 1, you will

- understand and describe how the Common Core State Standards for Mathematics (CCSSM) are organized; and
- understand and describe how the CCSSM are addressed in Investigations.

Overview of the CCSSM

Yes, the CCSSM are	No, the CCSSM are not
• • • • •	• • • • •

List the two types of standards in the CCSSM.

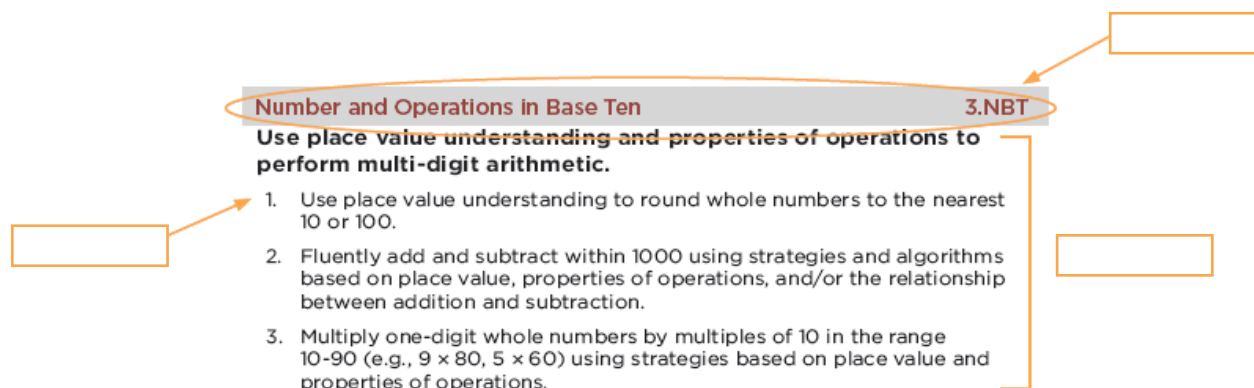
1. Standards for Mathematical _____
2. Standards for Mathematical _____

The Standards for Mathematical Content

Standards define what students should understand and be able to do.

Clusters are groups of related standards. Note that standards from different clusters may sometimes be closely related, because mathematics is a connected subject.

Domains are larger groups of related standards. Standards from different domains may sometimes be closely related.



(NGA Center and CCSSO 2010, 5)

The Standards for Mathematical Practice

1. Make sense of problems and persevere in solving them.
2. Reason abstractly and quantitatively.
3. Construct viable arguments and critique the reasoning of others.
4. Model with mathematics.
5. Use appropriate tools strategically.
6. Attend to precision.
7. Look for and make use of structure.
8. Look for and express regularity in repeated reasoning.

(NGA Center and CCSSO 2010, 6–8)

1. What student behaviors are associated with the practice?

2. How does Investigations address the mathematical practice? Provide an example to explain your response.

Solve a Higher-Level Thinking Task

Franco's daughter is 2 feet tall. Franco is 3 times as tall as his daughter. How tall is he?

- Explain what is happening in this problem.
- Use words, symbols, and pictures to compare Franco's height to his daughter's height.

C4)

Video Reflection: Phil Daro and the Standards for Mathematical Practice



Watch and Respond

What are the key points that Phil Daro talks about in the video?

How does Investigations address Phil Daro's key points?

Prepare for Next Session

Complete the following tasks before beginning Self-Paced Session 2.

Reading

- Read pages 8–9 in your *Investigations and the Common Core State Standards* Resource Book.
- Refer to this reading as you respond to the Discussion Board question.

Discussion Board

- Post your response to the following question:

After reading about the CCSSM on pages 8–9 in your *Investigations and the Common Core State Standards* Resource Book, list one concept that you found interesting, one concept that you learned, and one concept that you would like to know more about.

- Respond to two Discussion Board posts from other participants.

Final Assignment Preparation

- Choose an Investigations session for your final assignment Lesson Plan Implementation.

**Take some time to explore
the contents in your Core
Curriculum Package before
beginning Session 2!**

Session 2: A Closer Look at Investigations Curriculum Units

Session 2 Objectives

By the end of Session 2, you will

- implement the instructional components of a curriculum unit for optimal student learning and mathematical proficiency; and
- utilize Investigations resources in your classroom to personalize and differentiate instruction for your students.

Learn

Structure of a Curriculum Unit

Core Curriculum Package	The package includes all curriculum units, the <i>Implementing Investigations Guide</i> and the Resources Binder.
Curriculum Units	There are nine units per grade (but only seven units for Grade K): • <i>Mathematics in This Unit</i>: This section builds background in the mathematics content of each unit. It also provides an opportunity to look back to know how concepts are developed and look forward to see where they are going. The authors chunked the information by Math Emphases and Math Focus Points for easier reading and reference. • <i>Ongoing Assessment</i>: This includes tips to use while observing students at work and writing tasks that help you monitor progress in students' mathematical understanding. • Formal assessment opportunities: There are individual assessment activities and End-of-Unit Assessments. • <i>Daily Lesson Planner</i>: This includes the Math Focus Points, or teaching objectives and vocabulary for each session, the three to five parts of your sixty-minute session with time stamps, suggestions for grouping students during the session, and materials needed. • <i>Teacher Notes</i>: Written by the authors, the Teacher Notes support classroom instruction, classroom management, and assessment. The last Teacher Note of every unit provides support for having a discussion of students' solutions to the tasks in the End-of-Unit Assessment and support analyzing, interpreting, and scoring the students' work. • <i>Dialogue Boxes</i>: The set of sample field-tested classroom conversations convey how students typically express their mathematical ideas, what issues and confusions arise in their thinking, and how some teachers guide classroom discussions.

Session 2: A Closer Look at Investigations Curriculum Units

Implementing Investigations	These grade-level guides provide useful, ongoing support information. They are divided into the following sections: • <i>Part 1: Collaborating with the Authors</i>: Part 1 explains the guiding principles behind Investigations, how Investigations supports the teacher's role, and more. • <i>Part 2: Using Investigations</i>: This is a great place to start when beginning to implement Investigations. • <i>Part 3: Mathematics in Grade 3</i>: Part 3 summarizes the math behind Investigations. This information is also found in the appropriate curriculum unit front matter. • <i>Part 4: Classroom Routines</i>: This section summarizes each of the routines or Ten-Minute Math activities in the grade. • <i>Part 5: Technology in Investigations</i>: The rationale behind the use of technology in the Investigations curriculum is explained in Part 5. This section also has instructions for Shapes or LogoPaths software. • <i>Part 6: Professional Development</i>: A set of Teacher Notes that address topics and issues applicable to the curriculum as a whole is provided in this section. Each unit contains a set of Teacher Notes that offers information about the mathematical content of that unit and how students learn it. • <i>Part 7: Working with a Range of Learners</i>: Part 7 includes case studies written by the authors and field-test teachers. • <i>Part 8: Scope and Sequence/NCTM Curriculum Focal Points and Connections</i>: This section includes information about the strands and Math Emphases that are covered in each curriculum unit. • <i>Part 9: Math Terms and Index</i>: Part 9 is a comprehensive index for a particular grade level. Each curriculum unit book has a unit index in the back of the book.
Investigations and the Common Core State Standards Resource Book	The <i>Investigations and the Common Core State Standards</i> Resource Book includes everything needed to support teachers as they implement Investigations and the Common Core State Standards (CCSS) in their classrooms. Each grade-level book contains the following: • Snap-in and regular instructional plans for each curriculum unit, with a table for each Investigation that – lists all of the current and new sessions; – highlights where to adapt or omit existing content and/or add new content; and – identifies the mathematical practices and content standards that each session addresses. • All of the necessary teacher and student materials including – new activities and sessions; – new and adapted classroom routines/Ten-Minute Math activities ; – new Math/Teaching/Professional Development Notes; and – <i>Student Activity Book</i> pages. • Detailed correlations between the CCSS and Investigations
Differentiation and Intervention Guide	Available for Grades 1–5, these guides provide support for each Investigation, including • information about the important mathematics and help determining whether students understand, partially understand, or do not yet understand it; • an Extension, Practice, and Intervention activity, each with support for teaching English language learners; and • blackline masters, including an optional quiz.

Session 2: A Closer Look at Investigations Curriculum Units

Student Activity Book	The Student Activity Book provides students with a journal of their mathematical work. It contains the consumable pages for student work of an Investigation, recording sheets during a lesson, game directions and game sheets, daily practice pages, and homework pages. Icons at the top of each page show what type of page it is (for example, homework or daily practice pages). There is one page of practice for each daily class session. The pages that have a possibility of going home have a note on them to help guide parents. The book is also available in Spanish.
Student Math Handbook	This full-color hard-bound, nonconsumable book is available for Grades 1–5. It is also available in a 19" × 25" flip chart format for Grades K–2 and online for Grades K–5. Math Words and Ideas pages have a green band at the top and provide multiple representations of mathematical concepts and terms. Each page has the same format: the math concept as the title, a Word Bank, a few sentences about the concept, a visual representation of the concept, and a numerical representation and thinking question for the student as appropriate. The Math Words and Ideas pages are arranged by content strand (for example, Number and Operations, Geometry, Data). Game directions pages have an orange band on the top.
Manipulatives Kit	Each grade-level kit includes all of the student manipulatives, including overhead manipulatives, needed for the Investigations program.
Activity Cards	Each grade-level package includes the cards used in the games and activities. The cards are also available as individual sets.

Session 2 Pause and Reflect Question

Refer to the Professional Development pages listed for the session that you currently teach. Review the Math Focus Points for the session. How does the Professional Development relate to the Math Focus Points?

Grade and Unit: _____

Session: _____

Page: _____

What are the Math Focus Points for the session?

What does the Professional Development address?

How does the Professional Development relate to the Math Focus Points?

Differentiated Instruction

“The Common Core State Standards insist that all students must be supported in their learning so they can meet all of the standards. Differentiating instruction helps students achieve this goal.”

—Pearson Education, Inc. (2012a, T18)

“Differentiated instruction does not change what is taught; it changes how it is taught.”

—Basia Hall, Pearson Math Author
(Pearson Education, Inc. 2009)

How can Investigations resources support you in differentiating instruction?

Video Reflection Questions



Watch and Respond

After you have watched the video, respond to the following video reflection questions below.

- What expectations does the Investigation teacher have for all levels of learners in her classroom?
- What strategies does she use in her instruction to lead all learners toward mathematical proficiency?

Apply

Assignment: Supporting the Range of Learners

The purpose of this assignment is to learn about the individual students in your classroom so that you can identify strategies and curriculum resources that support their learning.

The following Investigation resources will help you fill in the third column: *Differentiation and Intervention Guide*, Professional Development case studies, Differentiation in this Unit, and the Differentiation feature in each session. The work that you do for this assignment will help you complete the final assignment.

Learner Subgroup	Number of Students in Subgroup	Describe two to three adaptations, instructional strategies, and/or curriculum resources that you can use to make mathematics more accessible and meaningful to this range of learners.
Gender	B: G:	
Intellectual Disabilities		
Advanced Learners/ Gifted		
Culturally/ Ethnically Diverse		
English Language Learner		
Struggling/ Reluctant Learner		

Prepare for Next Session

Complete the following tasks before beginning Self-Paced Session 3.

Reading

- Read the article, “Classroom Influences on Learning.”

Discussion Board

- Post your response to the following two questions:
 1. Think of the ways in which you can involve a range of students in activities and problems that are meaningful to them. Identify three specific activities, strategies, or resources that will support this.
 2. What are you presently doing in your math class to differentiate instruction? How will the Investigations resources support your current practices? What will you do differently?
- Respond to two Discussion Board posts from other participants.

Final Assignment Preparation

- Continue filling out the Lesson Plan Implementation organizer as you develop new understandings about implementing Investigations.

Classroom Influences on Learning

• The theory of constructivism suggests that we cannot teach students by telling. Rather, we must help them construct their own ideas using the ideas that they already own. This does not mean that we simply let students play around and hope that they will magically discover new mathematical ideas. On the contrary the manner in which you conduct your class plays an enormous role in what is learned and how well it is understood. Let's examine three factors that influence learning:

- student reflective thinking
- social interaction with other students in the classroom
- the use of models or tools for learning (manipulatives, symbolism, computer tools, drawings, and even oral language)

Each factor impacts what and how well students learn. Each one is significantly influenced by you, the classroom teacher.

Reflective Thought



Stop for a moment and see if you can come up with a good definition of reflective thinking. What does that phrase mean to you?

Whatever your description of reflective thinking, it almost certainly involves some form of mental activity. It is an active, not a passive, endeavor. You may have said that it involves figuring something out or trying to connect ideas in your head. You may have used the words “ponder” or “consider.” What you just did to try to come up with a definition of reflective thinking almost certainly involved reflective thinking.

If we assume that constructivist theory is correct, then we want students to be reflective about the ideas they need to learn. For a new idea you are teaching to be interconnected in a rich web of interrelated ideas, children must be mentally engaged. They must find the relevant ideas they possess and bring them to bear on the development of the new idea. In terms of the dots in Figure 1.1, we want to activate every gray dot a student has that is related to the new white dot we want them to learn. The more relevant gray dots used—the more reflective thinking—the better the new ideas will be constructed.

But we can't just hold up a big THINK sign and expect students to ponder the new thought. The challenge is to get them mentally engaged. As you will see later in

(Van de Walle and Lovin 2005, 4)

this chapter and throughout this book, the key to getting students to be reflective is to engage them in problems that force them to use their ideas as they search for solutions and create new ideas in the process. Two other related activities are also encouraged: writing about solutions to problems and having discussions with the rest of the class. Each is a method of promoting reflective thinking. Each should be built into most of your lessons.

Students Learning from Others

Reflective thought and, hence, learning are enhanced when the learner is engaged with others working on the same ideas. Students reside in classrooms. An interactive, thoughtful atmosphere in a classroom can provide some of the best opportunities for learning.

A worthwhile goal is to transform your classroom into what might be termed a “mathematical community of learners,” or an environment in which students interact with each other and with the teacher. In such an environment students share ideas and results, compare and evaluate strategies, challenge results, determine the validity of answers, and negotiate ideas on which all can agree. The rich interaction in such a classroom significantly raises the chances that productive reflective thinking about relevant mathematical ideas will happen.

The Interaction of Students’ Ideas with the Ideas of Others

Piaget helped us to focus on the cognitive activity of the child and to begin to understand how an individual uses ideas in a reflective manner to construct new knowledge and understanding. Vygotsky focused on social interaction as a key component in the development of knowledge. Vygotsky viewed the ideas that exist in the classroom, in books, and those shared by teachers and other authorities as distinct from the ideas constructed by the child. The well-formulated ideas that are external to the child he called *scientific concepts*, whereas those developed by the child (in the manner described by Piaget) he called *spontaneous concepts*.

Vygotsky talked about these two types of concepts as working in opposite directions, as shown in Figure 1.3. The scientific concepts work downward from external authority. As such, they impose their logic on the child. The spontaneous concepts bubble upward as a result of reflective activity. In Vygotsky’s *zone of proximal development*, the child is able to meaningfully work with the scientific concepts from outside. Here the child’s own conceptual understanding is sufficiently advanced to begin to take in the ideas from “above.”

It is not necessary to choose between a social constructivist theory that favors the views of Vygotsky and a cognitive constructivism that is built on the theories of Piaget (Cobb, 1996). In a classroom mathematical community of learners, students’ learning is enhanced by the reflective thought that social interaction promotes. At the same time, the value of the interaction for individual students is determined to a large extent by the ideas that each individual brings to the discussions. When, for any given child, the conversation of the classroom is within his or her zone of proximal development, the best social learning will occur.

Classroom discussion based on students’ own ideas and solutions to problems is absolutely “foundational to children’s learning” (Wood & Turner-Vorbeck, 2001, p. 186).

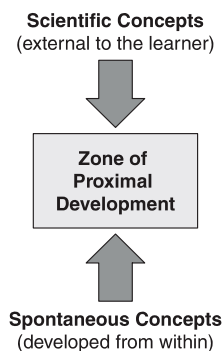


FIGURE 1.3 •••••

Vygotsky’s zone of proximal development is the place where new external ideas are accessible to the learner with those ideas already developed.

(Van de Walle and Lovin 2005, 5)

Mathematical Communities of Learners

In the wonderful book *Making Sense* (Hiebert et al., 1997), the authors describe four features of a productive classroom culture for mathematics in which students can learn from each other as well as from their own reflective activity.

1. Ideas are important, no matter whose ideas they are. Students can have their own ideas and share them with others. Similarly, they need to understand that they can also learn from the ideas that others have formulated. Learning mathematics is about coming to understand the ideas of the mathematical community.
2. Ideas must be shared with others in the class. Correspondingly, each student must respect the ideas of others and try to evaluate and make sense of them. Respect for the ideas shared by others is critical if real discussion is to take place.
3. Trust must be established with an understanding that it is okay to make mistakes. Students must come to realize that errors are an opportunity for growth as they are uncovered and explained. All students must trust that their ideas will be met with the same level of respect whether they are right or wrong. Without this trust, many ideas will never be shared.
4. Students must come to understand that mathematics makes sense. As a result of this simple truth, the correctness or validity of results resides in the mathematics itself. There is no need for the teacher or other authority to provide judgment of student answers. In fact, when teachers routinely respond with “Yes, that’s correct,” or “No, that’s wrong,” students will stop trying to make sense of ideas in the classroom and discussion and learning will be curtailed.

Classrooms with these characteristics do not just happen. The teacher is responsible for creating this climate. It happens over time in two ways. First, there must be some direct discussion of the ground rules for classroom discussions. Second, teachers can model the type of questioning and interaction that they would like to see from their students.

(Van de Walle and Lovin 2005, 6)

Session 3: Building a Mathematical Community of Learners

Session 3 Objectives

By the end of Session 3, you will

- orchestrate meaningful mathematical discussions that build and support a community of learners; and
- explain how Investigations games enhance learning and increase understanding.

Learn

Mathematical Discussions

The Math Focus Points for Discussion and sample teacher dialogue found in the text for every discussion will help you guide the discussion. In preparing for class, ask yourself the following questions:

- What do I want the discussion to accomplish?
- What do I want all students to take away from this discussion?
- How will the time spent as a whole class enhance the work that students have done individually or in pairs or groups?

Play a Game

Name _____

Date _____

Number Puzzles and Multiple Towers



Division Compare

You need

- Compare Cards (1 deck per pair)

Play with a partner.

- 1 Divide the deck of cards evenly so that each player has the same number of cards.
- 2 Each player turns over two cards.
- 3 Using the larger number as the dividend and the smaller number as the divisor, make a division problem. (For example, if your cards are 80 and 700, the division problem is $700 \div 80$.)
- 4 Estimate, compare, and reason about the relationships of the numbers to figure out which player has the greater quotient (answer). Discuss how you know which answer is greater.
- 5 The person with the greater quotient takes all of the cards. If the quotients are equal, players turn over two new cards and the person with the greater quotient takes all of the cards.
- 6 Play for a given amount of time or until one player has all of the cards.

Variation

You need

- Digit Cards (1 deck per pair)

Play the same game using Digit Cards with the "0" cards removed. Each player draws five cards. Using the cards in the order they were picked, choose the first three cards to form the dividend and the last two cards to form the divisor. (For example, if you picked 8, 1, 5, 9, 4, your division problem would be $815 \div 94$.)

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M52 Unit 1

Sessions 3.6 and 3.7

Name _____

Date _____

Number Puzzles and Multiple Towers



Division Compare Recording Sheet

After you have played a few rounds of *Division Compare*, complete this sheet.

Place a $<$, $>$, or $=$ in the box between the problems.

1. Your problem: _____ Partner's problem: _____

_____ $\div$ _____ _____ $\div$ _____

How did you decide whose problem has the greater quotient? Explain your reasoning.

2. Your problem: _____ Partner's problem: _____

_____ $\div$ _____ _____ $\div$ _____

How did you decide whose problem has the greater quotient? Explain your reasoning.

3. Your problem: _____ Partner's problem: _____

_____ $\div$ _____ _____ $\div$ _____

How did you decide whose problem has the greater quotient? Explain your reasoning.

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Sessions 3.6 and 3.7

Unit 1

M53

Name _____

Date _____

Number Puzzles and Multiple Towers

Compare Cards (page 1 of 2)



2	3	4	5
<u>6</u>	7	8	<u>9</u>
10	20	30	40
50	60	70	80
90	10	20	30

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M44 Unit 1

Sessions 2.3–2.5

Name _____

Date _____

Number Puzzles and Multiple Towers

Compare Cards (page 2 of 2)



40	50	60	70
80	90	100	200
300	400	500	600
700	800	900	400
500	600	700	800

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Sessions 2.3–2.5

Unit 1

M45

Pause and Reflect

Reflect on the game that you played.

What type of discourse should you do with students after playing the game?

What two specific questions can you ask to assess students' understanding of the mathematics in the game and prompt mathematical discourse?

Computational Fluency

Math Games 3–5



Close to 100

You need

- Digit Cards (deck of 44)
- *Close to 100* Recording Sheet for each player

Play alone, with a partner, or in a small group.

- 1 Deal out six Digit Cards to each player.
- 2 Use any four cards to make two numbers; for example, 6 and 5 could make either 56 or 65. Wild cards can be used as any numeral. Try to make numbers that, when added, give you a total that is close to 100.
- 3 Write these two numbers and their total on the *Close to 100* Recording Sheet; for example, $42 + 56 = 98$.
- 4 Find your score. Your score is the difference between your total and 100. For example, if your total is 98, your score is 2. If your total is 105, your score is 5.
- 5 Put the cards you used in a discard pile. Keep the two cards you did not use for the next round.
- 6 For the next round, deal four new cards to each player. Make more numbers that come close to 100. When you run out of cards, shuffle the discard pile and use those cards again.
- 7 Five rounds make one game. Total your scores for the five rounds. The player with the LOWEST score wins.

G6

In Practice



Watch and Respond

What are some strategies and ideas that you can use in your own classroom to organize game activities and use them to build mathematical proficiency?

Apply

Setting Up the Mathematical Community

Helping Students Prepare to Share Ideas

The sharing of ideas and strategies during math discussions is an important component of a productive math community. However, as second-grade teacher Betty Jane O'Donnell discovers, many students need help figuring out just what to share during whole-group discussions. In this case, Ms. O'Donnell reflects on how she helps her students prepare for math discussions by raising specific questions for them to focus on as they work on activities prior to the discussion.

The first month of the school year is when everything is possible. It is also the month for introducing routines and fostering habits of work that will transform the possibility of the first month into the satisfaction of the last. At the start of each school year, I gather the students together with the goal of generating a list of how we work together during math time. Their ideas usually include statements like the following:

"We use materials and supplies."

"We think and explore."

"We clean up."

"We share ideas."

It was a lovely list, but I wasn't convinced that everyone was actively engaged in bringing such ideas to life, or whether my students understood the extent of what they said. I explained that from time to time I would choose an idea from the list, and we would talk about what it looks like in our math class.

After talking about the easier items on the list, such as materials and cleanup procedures, one day I asked students to think about how they share ideas. I wanted them to get a picture in their heads of a time when they shared ideas in math. Here is part of that conversation:

Kaitlyn: We share ideas at the end of Math Workshop.

[There was a long pause. I waited.]

Lacey: It's fun to get to know each other better.

[Another long pause followed.]

Michaela: We usually sit in a circle so that we can see each other and share ideas.

Teacher: So what are some of the ideas that we've shared lately? What have we been learning about in math?

Bodies started shifting. A general sense of unease was spreading around the circle. I decided to bring the class back to the specifics of what they had been doing over the past several days.

Teacher: This seems to be kind of a tricky question. Let me say it in a different way. What have we been doing in math?

David: Card games.

Teacher: Yes, we've been playing card games. And what have those games been about?

Sherry: Making tens.

Hannah: We add the numbers up.

Teacher: We've been adding the numbers on the cards. Today will be the last day that we will play those card games for a while. What are you going to be thinking about when you play those games?

The blank looks and long pause told me that my second graders were having a hard time making sense of what I was asking. I reminded myself how important it is to be explicit to help students focus on the mathematical ideas. Especially this early in the year, once I have identified the central mathematics in their activities, I need to share this focus with my students.

Teacher: As you play the games, I would like you to think how you know what card you need in each game. How do you know what card you need when you play *Tens Go Fish*? How do you know what card to pick to play *Make 10*? That is what we will be talking about when we share our ideas.

As my students set off to Math Workshop, I thought about my role in helping them participate in our discussions and making their participation meaningful. By explicitly articulating questions about the math focus, I help all students prepare to share their ideas.

In this case, Ms. O'Donnell reflects on her role in helping her students have successful math conversations. She realizes that "What are you going to be thinking about when you play those games?" was too vague. She decides that for all students to have a point of entry during sharing time, she needs to ask clearer and more explicit questions so that her students can focus on the math ideas embedded in the games. Helping students with this focus is a first step toward deeper math conversations.

Questions for Discussion

1. What information did Ms. O'Donnell provide to her students before they began the games *Tens Go Fish* and *Make 10*? How did she hope this information would help them prepare for the math discussion to follow?
2. What are the different purposes math discussions serve in your classroom? How do you make the focus of a particular math discussion explicit for your students?
3. What strategies do you or could you use to help students prepare in advance for math discussions?

Assignment: Making Math Discussions Explicit

Choose a game or discussion activity from the Investigations session that you are using for your final assignment. Identify the learning goal of the activity, and decide what mathematical discourse you want to occur. Think about the strategies that you will use to help students prepare in advance for this math discussion.

Describe the Investigations activity.

What is the learning goal of the activity? What will students be able to do after they complete the activity?

Before: What mathematical discussions are you hoping will occur during the activity? How will you prepare your students in advance for this math discussion?

During: Describe the anticipated student responses during the Investigations activity. How will you facilitate their math discussions to make the learning goal explicit? What questions will you ask to extend their thinking?

After: What will you listen for as evidence of learning from the Investigations activity? Will each group of students present what they learned?

Prepare for Next Session

Complete the following tasks before beginning Self-Paced Session 4.

Reading

- Read the article, “Assessing to Learn and Learning to Assess” on the next page.

Discussion Board

- Post your response to the following two questions:
 1. What type of discourse should you do with students after games to make the learning explicit and meaningful?
 2. How can you help the range of learners both participate and learn from each other during class discussions? What strategies do you use to do this in your classroom?
- Respond to two Discussion Board posts from other participants.

Final Assignment Preparation

- Continue filling out the Lesson Plan Implementation organizer as you develop new understandings about implementing Investigations.

Assessing to Learn and Learning to Assess

by Cathy Seeley, NCTM President 2004–2006
[*NCTM News Bulletin*, September 2005](#)



With the public's attention focused on how well our schools are educating children in mathematics, students face assessment on a regular basis. At the classroom level, teachers use a variety of assessment measures and strategies to determine whether their students are learning the mathematics they are supposed to learn. Schools and school districts often administer tests across classrooms to determine whether students have reached a certain level of achievement or whether they are meeting particular benchmarks. Every state also administers large-scale accountability tests as required by law. Is all this testing too much, or does it help students learn? When we *assess to learn*, we seek information that allows teachers to find out what students know in order to improve their mathematics learning. This means *learning to assess* all the kinds of mathematical knowledge we identify as important. By assessing to learn and learning to assess, we can maximize the positive impact of assessment on students' learning without unnecessary negative consequences.

What Is Assessment For?

When a teacher wants to know whether a student or group of students is learning what is expected on a day-to-day basis, the teacher may use a variety of measures such as quizzes, interviews, projects, tests, or even purposeful conversation. If a test is used to determine the breadth of a student's mathematical knowledge and level of thinking, it is likely to include opportunities for the student to produce extended responses that demonstrate a thought process in addition to measuring a range of mathematical content and skills. To measure a program's effectiveness for a large number of students, a test needs to be efficient to administer and quick and economical to score. Each of these purposes calls for a different type of measure, with a specific format, scope, and context for administration.

Regardless of the purpose of an assessment, one factor is crucial to all assessments. They must be aligned with the particular mathematics that students are expected to learn. Ideally, this alignment will be evident in the content of the test or assessment and supported by the assessment format and manner in which the results are interpreted and used.

What Does Assessment Tell Us?

Teachers and students can benefit from assessment results that tell what a student knows and that identify a student's potential misunderstandings. The most useful assessment results for directly influencing students' learning are those that are immediate and specific. When we acknowledge what students are doing well and adjust or guide students as they first develop misunderstandings, they are far more likely to learn mathematics correctly and have that learning last. An effective teacher knows that it is important to measure students' understanding of mathematical concepts and ideas, evaluate their proficiency in skills, and give them the opportunity to apply what they have learned to a variety of situations beyond the immediate

context in which the mathematics was learned. Results of large-scale accountability tests should be interpreted and used carefully only for the purposes for which they were intended. For making day-to-day decisions, the teacher is the best person to assess mathematics learning, and the classroom is the best context in which to do so.

Too Much Testing?

Ideally, assessment should be seamlessly woven into the fabric of teaching and learning, minimizing interruptions in instructional time and maximizing the immediate impact on students' learning. When assessment is part of the learning process, it does not need to sidetrack an effective mathematics program.

The bottom line is that the purpose of any mathematics assessment must be to improve students' learning. When an assessment measure is well aligned with—and integrated into—the system of mathematics teaching and learning, preparing students to perform well should involve little more than teaching the mathematics program well.

Learning to Assess

Developers of accountability assessments can improve assessments by incorporating problem solving, open-ended items, and problems that assess understanding as well as skills. As consumers of test data, teachers, supervisors, administrators, and families can learn what test data do and do not tell us. In our own classrooms, we can refine our skills so that we design assessment measures that clearly show what students know, with assessment integrated as part of the teaching and learning process.

Session 4: Assessment in Investigations

Session 4 Objectives

By the end of Session 4, you will

- describe assessment features in Investigations that will target instructional goals; and
- use assessment results to inform instruction.

Learn

Video Reflection

As you watch the video, reflect on the following questions:

How do the students perceive this assessment?

What are the benefits of using embedded assessment as opportunities for learning?

Pause and Reflect

1. Select one of the benchmarks that you observed in your curriculum unit.
2. Look at the lessons listed in Key Activities to Observe. In what part of the lesson could you assess this Benchmark?
3. Where else in the unit does the benchmark assess? How does the benchmark assess it?
4. Notice that some of the assessments have a checklist icon in the table. Look at the Session sidebar to see how the checklist is organized. How could you use this during instruction?



Watch and Respond

Once you have watched the video about Ongoing Assessment, respond to the reflection questions that follow.

- What strategies and methods did the teacher use to assess students during instruction?
- What did she look for as evidence of understanding the mathematics?

Apply

Assignment: Assessing Students' Understanding

Collect a sample of students' work on a selected task from the Investigations session that you are using for your final assignment. Use the check sheet below to assess students' understanding of the mathematical concepts and make instructional decisions.

You will determine if each student

- understands the mathematics;
- partially understands the mathematics; or
- does not understand the mathematics.

Description of the task or assessment:
Mathematics that the student needs to learn:

	✓ Understands the Mathematics?	✓ Partially Understands the Mathematics?	✓ Does Not Understand the Mathematics?
Indicators/Evidence to Look For			
Student			

Prepare for Next Session

Complete the following tasks before live Session 5.

Discussion Board

- Post your response to the following question:
What does assessment look like in your classroom, and how will it change based on what you have learned in this session?
- Respond to two Discussion Board posts from other participants.

Final Assignment Completion

- Teach the Investigations lesson that you have been focusing on in your final assignment.
- Complete the final assignment by filling in all parts of the Lesson Plan Implementation graphic organizer and uploading it to the dropbox for this course on Pearson LearningStudio before Session 5.
- Consider possible topics of discussion that relate to your Investigations Lesson Plan Implementation to share during a Lesson Study group discussion during the next live session.

Session 5: Planning an Investigations Unit

Session 5 Objectives

By the end of Session 5, you will

- identify program features for effective unit planning; and
- debrief an Investigations lesson in a collaborative setting.

Unit Planning

- When planning an Investigations unit, what is most important to you?
- What additional instructional resources do you like to include? (Examples include technology, daily written formative assessments, journaling, skill practice, daily vocabulary practice, and so on.)

Investigations Lesson Debrief

Share one aspect of your Investigations lesson that you taught as a part of the final assignment. The list below includes possible suggestions on what you might want to share.

- Share a success.
- Share something that you will do differently next time.
- Share something that you learned from your students.
- Share how you either
 - addressed the mathematical practices;
 - differentiated instruction;
 - orchestrated a purposeful discussion; or
 - assessed students' understanding.

Peer Feedback

As a group participant during the breakout presentations, you will do the following:

- Note positive feedback regarding the Lesson Plan Implementation.
- Ask any questions regarding the implementation of Investigations features.

Closing

3 . . . 2 . . . 1 Reflection

List **three** things that you learned during this course.

1. _____
2. _____
3. _____

Write **two** strategies that you learned today that you would like to incorporate into your classroom.

1. _____
2. _____

Share **one** question that you have about the CCSS.

1. _____

Discussion Board

Respond to the Session 5 discussion questions.

- What is one question that you have about the CCSS?
- What is the first thing that you will do in your classroom to prepare your students to be more mathematically proficient?

Appendix

Lesson Plan Implementation

During this course, you will learn essential steps for implementing Investigations for the Common Core in your classroom. This activity will help you plan a learning environment that prepares your students to be mathematical thinkers and problem solvers. Use the resources embedded in your instructional program to guide you as you organize for instruction.

Directions: Use the organizer to plan and teach an Investigations lesson. Evaluate the lesson implementation by completing the reflective questions.

Unit Title:	
Investigation:	
Session:	
Summary of Session/Math Focus Points:	

Appendix

CCSSM
What Common Core adaptations does the Session recommend?
How does the session address the Standards for Mathematical Practice?
Describe a student behavior that you observed during the session that showed evidence of applying one or more of the Standards for Mathematical Practice.

Differentiated Instruction
Look in your Differentiation and Intervention Guide, the Differentiation in this Unit page in your Curriculum Unit, and the Differentiation support in the session. Which of these resources will be most useful to you in this session?
Describe the types of learners in your classroom. How did you differentiate instruction during the session?

Appendix

Mathematical Discussions

Summarize the opportunities for mathematical discussions in the session. What mathematical conversations do you hope to develop?

Describe how mathematical discussions developed during the session. Did the students show evidence of thinking and reasoning in their discussion?

Assessment

What opportunities for assessment are in the session?

Describe how you assessed student understanding during the session. How did it inform your instruction?

Rubric

**Investigations in Number, Data, and Space® © 2012 for the Common Core
Product Implementation Essentials Fused Online Course
Final Assignment Rubric**

The participant filled in each area of the Lesson Plan Implementation chart with appropriate information.	Yes ☐ No ☐	Comments:
The Lesson Plan Implementation describes how the participant addressed the Common Core State Standards for Mathematics.	Yes ☐ No ☐	Comments:
The Lesson Plan Implementation describes opportunities in the lesson to differentiate instruction.	Yes ☐ No ☐	Comments:
The Lesson Plan Implementation describes how mathematical discussions developed during the lesson.	Yes ☐ No ☐	Comments:
The Lesson Plan Implementation includes opportunities for assessment.	Yes ☐ No ☐	Comments:

Additional Comments:

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